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Special Session: New Applications of Evolutionary Linguistics

Group learning Integrating Behaviors and Emotions recognition to Assess Student Engagement in Classroom Context

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Abstract—Group-based learning has become a cornerstone of modern education, fostering collaboration, critical thinking, and enhanced problem-solving skills among students. In this study, a deep learning framework, called *MoGL-Net*, is proposed. This framework integrates skeleton-based posture, sequential action features, facial emotion and classroom context to assess student engagement. Besides, a fusion strategy employing Self-Attention and Cross-Attention mechanisms is utilized to capture interdependencies between them. Additionally, this study also introduces a real-world dataset, *HCMUE-SEGL*, consisting of 1,663 video segments collected from 61 students participating in small-group learning activities. Experiments results show that *MoGL-Net* model achieves an accuracy of 90.70% on *HCMUE-SEGL*, highlighting the potential of multimodal learning for developing intelligent systems to monitor and objective classroom analytics in group learning environments.

Index Terms—Group learning, behavior recognition, Facial expression recognition, Classroom context

I. INTRODUCTION

In the current educational context, the teaching goal is to promote and comprehensively develop the qualities and competencies of learners. Accordingly, active teaching methods are encouraged to foster the initiative and engagement of students in learning tasks [1], [2]. Among these methods, group-based learning has been widely applied due to its ease of organization and superior effectiveness in enhancing educational quality. For these collaborative learning activities to be successful, the active and positive participation of students is essential [3], [4]. Therefore, teachers need to observe and assess the level of student participation in collaborative learning activities to provide timely adjustments and support. However, traditional methods such as teacher observations or self-assessments are often subjective and time-consuming. Thus, there is a need for an automated, objective evaluation framework that can analyze non-verbal signals such as behavior and facial expressions to provide timely and effective feedback [5], [6].

With the development of deep learning, many studies have applied it to assess student engagement in group learning. These models typically provide assessments based on the analysis of behavioral signals [7]–[9] or facial expressions [10], [11]. However, these methods have not fully utilized the relationship and interaction between behavior and emotions,

leading to an incomplete or inaccurate reflection of the actual engagement levels of learners [12]. To address this, multimodal methods are potential. They combine multiple information streams, including both behavioral and emotional features [13], [14]. However, most of these approaches still focus on analyzing individual behavior-emotion in traditional classroom settings and do not account for the contextual features of interactions between group members in group learning—an essential factor for reflecting group-level engagement.

This study proposes a multimodal deep learning model to assess individual student engagement in collaborative group learning environments, called *MoGL-Net*. The model integrates skeletal pose graphs, full-body RGB, facial expressions, and group context, using a two-stage attention mechanism to classify engagement into four levels: Disengaged, Normal, Engaged, and Very Engaged. Besides, a real-world multimodal dataset, the *HCMUE Student Engagement in Group Learning Dataset (HCMUE-SEGL)*, comprising 61 students with 1,663 annotated samples was constructed, capturing natural group learning interactions with synchronized pose, RGB, face, and context data. Experimental results demonstrated that integrating multimodal features significantly improves engagement classification accuracy, validating the effectiveness of the proposed approach.

II. RELATED WORKS

The assessment of learner engagement through artificial intelligence applications has become a significant area of research in recent years, especially within the growing context of digital classrooms and collaborative learning. One of the most common approaches is the analysis of behavioral cues through body posture and physical movements, observable via deep learning methods. The study in [8] proposed fine-tuning the C3D (3D CNN) model to learn spatio-temporal features of micro-body gestures to evaluate engagement levels in online learning environments. Despite achieving promising results, the absence of emotional analysis leads to an incomplete assessment of engagement. This is because a learner might appear engaged physically while actually feeling bored or anxious, which can negatively impact learning.

Therefore, facial expression analysis is emerging as a parallel research direction for inferring emotional states associated with engagement [15], [16]. The study in al. [16] developed a system to evaluate learner engagement in a classroom setting by analyzing facial expressions and head posture. Nevertheless, much like approaches that rely solely on behavior, a singular focus on facial expressions may fail to exploit the connection between behavior and emotion, which is essential for assessing the learner’s true engagement level.

Moreover, some studies have proposed multimodal approaches for assessing engagement. The studies in [17], [18] introduce models for real-time engagement monitoring in traditional classrooms by combining three modalities: facial expressions, head posture, and head movement. However, these methods exhibit shortcomings as they are designed exclusively for traditional classroom settings where learners work individually and thus ignore group-level interactions. Recent advances in deep learning for image analysis also emphasize architecture optimization and attention-based decoding strategies [19], but these remain confined to static image-level prediction and cannot capture the temporal, affective, and social dimensions that characterize real classroom interactions. In contrast, our method employs a hierarchical self- and cross-attention framework that fuses individual signals with classroom context. The context branch enables the model to capture interaction dynamics within the group, rather than treating engagement as an isolated individual phenomenon. By grounding individual signals in group-level context, our approach overcomes the limitations of unimodal or static methods and establishes a comprehensive foundation for engagement recognition in collaborative learning environments.

Although initial studies have contributed data on learner engagement [20], [21], these contributions are often unimodal, focusing primarily on a single aspect like behavior or emotion. To address this limitation, subsequent research has focused on constructing multimodal datasets [22], [23]. However, these datasets have predominantly concentrated on online learning environments. In this study, a new multimodal dataset for learner engagement in group activities is also constructed in both behavioral and emotional cues.

III. DATASET CONSTRUCTION

This section presents the building of a new multimodal dataset for assessing student engagement in group learning, called HCMUE-SEGL. This dataset is categorized into four levels: Very Engaged, Engaged, Normal, and Disengaged. These levels are constructed based on pairs of behavioral and emotional expressions as detailed in Table I.

The dataset includes nine types of behaviors (Standing, Raising hand, Discussing, Writing, Using Phone, Reading, Sleeping, Looking Around, Boring) and five basic emotions in a classroom environment (Surprise, Angry, Happy, Neutral, Sad). In which, the process of defining and grouping these behavior-emotion pairs into corresponding engagement levels was based on a survey conducted with 61 volunteers, the majority of whom were experienced high school teachers and university lecturers familiar with applying group-based learning methods. The volunteers were asked to select the most appropriate engagement level for each pair of expressions, and the level with the highest number of votes was chosen as the official label.

TABLE I: Description of Behavior and Emotion by Level of Engagement

Level of Engagement	Description of Behavior and Emotion
Very Engaged	Standing with an angry face, Standing with a happy face, Discussing with a surprised face, Discussing with an angry face, Discussing with a happy face, Raising hand with an angry face, Raising hand with a happy face, Writing with an angry face.
Engaged	Standing with a sad face, Standing with a neutral face, Raising hand with a sad face, Raising hand with a neutral face, Discussing with a sad face, Discussing with a neutral face, Writing with a happy face, Writing with a neutral face, Reading with an angry face, Reading with a happy face, Reading with a surprised face, Using phone with a happy face, Using phone with a surprised face, Using phone with an angry face.
Normal	Using phone with a neutral face, Using phone with a sad face, Reading with a sad face, Reading with a neutral face, Writing with a sad face.
Disengaged	Sleeping with a neutral face, Looking around with a neutral face, Boring with a neutral face.

The data collection process was conducted in various classrooms and at different times to increase the contextual diversity of the dataset. Recording over multiple days helped the learners’ emotions to be expressed more naturally, thereby enhancing the authenticity and objectivity of the data collected in the study. The dataset features the participation of 61 volunteers, with an average age of 21, as shown in Table II.

TABLE II: Demographic information of participants

Quantity	Age	Gender	Education level	Nationality
3	19	2 Male, 1 Female	Undergrad student	Vietnam
11	20	6 Male, 5 Female	Undergrad student	Vietnam
35	21	30 Male, 5 Female	Undergrad student	Vietnam
5	22	4 Male, 1 Female	Undergrad student	Vietnam
6	23	3 Male, 3 Female	Graduate student	Vietnam
1	26	1 Male	Undergrad student	Vietnam
Total: 61				

To protect privacy and personal information, all volunteers were thoroughly briefed on the procedures and proactively signed consent forms permitting the use of their images for research purposes. The cameras used for recording have a resolution of 720p and a frame rate of 30fps. The cameras were placed at the four corners of the table, as illustrated in Figure 1. This setup helps to clearly capture the learners’ behaviors and emotions from different perspectives.

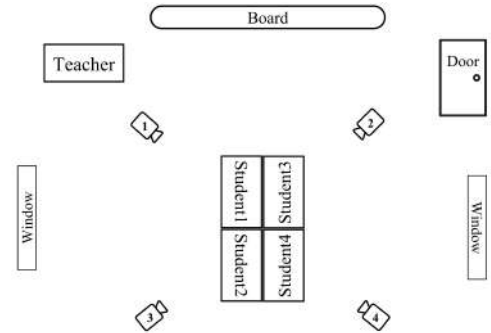


Fig. 1: Simulated diagram of volunteer seating arrangement and camera positions during filming.

The collected raw videos are first segmented into short

5-second clips. Next, these clips are processed through YOLOv11 [24] combined with the BoT-SORT algorithm to track each individual. From the frame sequence of each tracked subject, we manually filter and select 15 equally spaced frames. In cases where the subject's face is not clearly visible in a selected frame, we replace it with the nearest frame where the face is clear. The labeling process was carried out by 11 volunteers, with each data sample independently annotated by five individuals with teaching experience. Based on the analysis of the behavioral-emotional pairs in the 15-frame sequence, the annotators assigned an engagement level to each sample. The final label was determined through majority voting, which helps mitigate individual subjectivity. In addition, samples with inconsistent votes were re-checked and resolved through consensus, thereby enhancing annotation reliability and consistency.

Subsequently, this 15-frame sequence continues to be processed for feature extraction. The YOLO-Pose [25] framework is used to extract skeleton data, and RetinaFace [26] is used to detect and crop the facial region. Besides, the middle frame of the 15-frame sequence is extracted to represent the classroom context at that moment. After processing, the total number of collected samples is 1663. Detailed statistics of the samples are presented in Table III.

TABLE III: Statistics of the number of samples in the HCMUE-SEGL dataset

Category	Very Engaged	Engaged	Normal	Disengaged	Total
Number of samples	398	428	331	506	1663

IV. A MULTIMODAL NETWORK FOR GROUP LEARNING ENGAGEMENT ASSESSMENT - MOGLE-NET

This section presents MoGLE-Net, a deep learning model for estimating student engagement in group learning by integrating behavioral, emotional, and contextual features, as illustrated in Figure 2. The model processes 15-frame video sequences, where each branch contributes unique information: the Pose Skeleton branch, using SKE-A3TGCN [27], captures body posture and motion to reflect physical engagement; the Action RGB branch, based on ResNet50-LSTM, analyzes full-body actions to track active or passive participation; the Facial Emotion branch, using ViViT [28], extracts emotional expressions to gauge emotional engagement; and the Group Context branch, using ViT [29], captures spatial dynamics to incorporate group-level interaction.

A. Multimodal Feature Extraction

1) Pose-Based Behavioral Representation

The skeleton sequence $\mathcal{G} = \{G_t \in \mathbb{R}^{13 \times 2} \mid t = 1, \dots, T\}$, with $T = 15$, represents a 13-joint skeleton (Nose, Left Eye, Right Eye, Left Ear, Right Ear, Left Shoulder, Right Shoulder, Left Elbow, Right Elbow, Left Wrist, Right Wrist, Left Hip, Right Hip) per frame. Each frame $G_t = (V_t, E_t)$ is a graph where nodes $v \in V_t$ contain 2D coordinates for joint position and edges $e \in E_t$ encode distance and angles, consistent with the vertex and edge feature representations in the original paper [27]. The SKE-A3TGCN model processes $\mathcal{G}$ via spatial graph convolution:

$$H_t = \sigma \left(\hat{D}^{-1/2} \cdot \hat{A} \cdot \hat{D}^{-1/2} \cdot X_t \cdot W \right) \quad (1)$$

where X_t is the node feature matrix at time t , $\hat{D}^{-1/2}$ is the inverse square root of the degree matrix $\hat{D}$, used to normalize the adjacency matrix $\hat{A}$, ensuring balanced influence from nodes with different degrees, and W is a learnable weight matrix. The nonlinear activation function $\sigma(\cdot)$, typically ReLU, is applied after matrix multiplication to improve the model's ability to capture complex patterns. The resulting output, H_t , is the feature matrix of the graph convolution layer, containing the spatial features aggregated from neighboring nodes, representing the spatial relationships between joints at time step t . Temporal dynamics are captured via attention:

$$\alpha_t = \text{softmax}(q^\top \tanh(W_h H_t + b_h)) \quad (2)$$

$$\mathbf{F}_G = \sum_{t=1}^T \alpha_t H_t \in \mathbb{R}^{512} \quad (3)$$

where W_h, b_h are learnable parameters, q is a query vector, and α_t weights frame contributions, following the temporal attention mechanism. The embedding $\mathbf{F}_G$ captures motion and pose trends for engagement prediction, leveraging vertex and edge features.

2) Full-Body Action Encoding

The RGB action sequence $\mathcal{I} = \{I_t \in \mathbb{R}^{224 \times 224 \times 3} \mid t = 1, \dots, T\}$ is processed using a ResNet50-LSTM architecture to capture spatial and temporal motion patterns. Each frame I_t is passed through a ResNet50 backbone, extracting spatial features $f_t \in \mathbb{R}^{2048}$. The sequence $\{f_1, \dots, f_T\}$ is fed into a uni-directional Long Short-Term Memory (LSTM) network to model temporal dependencies:

$$h_t = \text{LSTM}(f_t, h_{t-1}) \quad (4)$$

where h_t is the hidden state at time step t . The output of the full-body action encoding branch is $\mathbf{F}_B \in \mathbb{R}^{512}$, which encapsulates dynamic full-body behavioral cues, leveraging spatio-temporal visual feature. This encoding enhances engagement recognition by integrating appearance-based motion information, thus providing a comprehensive understanding of the subject's physical involvement in the group learning activity.

3) Spatial-Temporal Facial Emotion Encoding

The facial expression sequence $\mathcal{F} = \{F_t \in \mathbb{R}^{224 \times 224 \times 3} \mid t = 1, \dots, T\}$, with $T = 15$ frame, is processed using the ViViT model to capture spatial-temporal facial features. Each frame F_t is cropped using RetinaFace, and the resulting face images are split into 3D tubelets, with each tubelet consisting of two consecutive frames and 16x16 pixel patches. These tubelets are projected into embeddings and passed through multiple Transformer layers.

The model applies Spatial Attention to capture relationships between patches within each frame and Temporal Attention to model expression dynamics over time. The embeddings are then aggregated via mean pooling across both spatial and temporal dimensions to produce the final facial emotion embedding $\mathbf{F}_E \in \mathbb{R}^{512}$. This embedding captures emotional intensity and variation, contributing to the engagement prediction by integrating dynamic facial expression information.

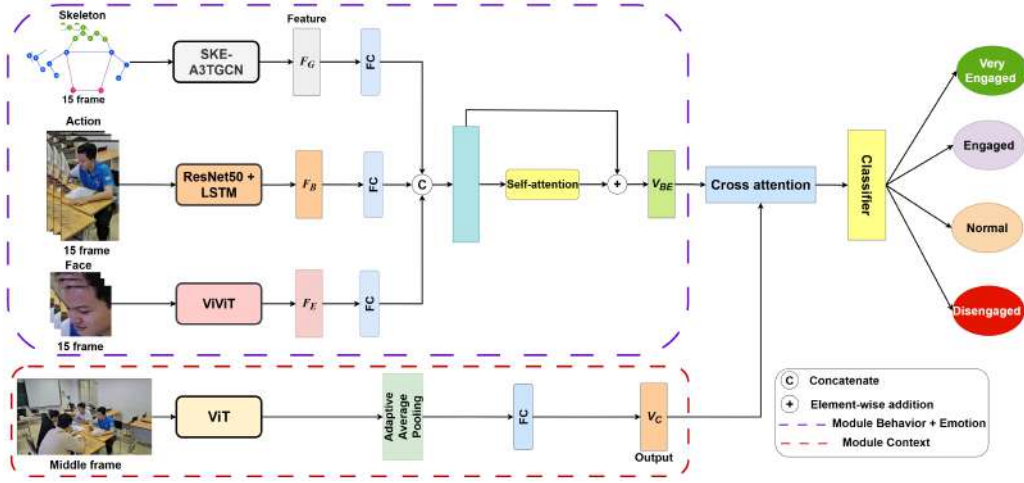


Fig. 2: The architecture of MoGLE-Net model.

4) Contextual feature

The context module processes a middle frame $C \in \mathbb{R}^{224 \times 224 \times 3}$ using the pre-trained ViT backbone. Unlike CNNs, ViT uses self-attention to capture global relationships across the entire frame, allowing it to better model group-level interactions and spatial context. This ability to focus on relevant areas, regardless of their spatial distance, enhances the understanding of group interactions in classroom settings.

The final context embedding $\mathbf{F}_C \in \mathbb{R}^{512}$ is fused with behavioral and emotional features through cross-attention to improve engagement prediction.

B. Hierarchical Attention-Based Multimodal Fusion

To integrate unimodal embeddings from skeleton ($\mathbf{F}_G$), action ($\mathbf{F}_B$), and facial emotion ($\mathbf{F}_E$) branches, we employ a hierarchical attention mechanism for engagement prediction. First, the unimodal embeddings are concatenated to form the behavior-emotion token set:

$$\mathbf{T}_{BE} = [\mathbf{F}_G; \mathbf{F}_B; \mathbf{F}_E] \quad (5)$$

$$\mathbf{A}_{BE} = \text{SelfAttention}(\mathbf{T}_{BE}) \quad (6)$$

The self-attention mechanism is particularly effective when applied to features from the same data source, as it allows the model to learn internal dependencies between them. In this case, the pose, full-body, and facial emotion features are all expressions of the same individual's engagement, so their relationships are best learned within the same context. By applying self-attention, the model dynamically adjusts the importance of each modality based on its relevance at any given time, allowing for a more accurate representation of the person's engagement. After applying self-attention, the raw features are combined with the attended features through element-wise addition, followed by mean pooling across the tokens:

$$\mathbf{v}_{BE} = \text{MeanPool}(\mathbf{T}_{BE} + \mathbf{A}_{BE}) \quad (7)$$

This fusion process ensures that both raw and attended features contribute to the final behavior-emotion vector $\mathbf{v}_{BE}$,

which effectively captures the subject's behavioral and emotional state.

Group context is incorporated via cross-attention, with $\mathbf{v}_{BE}$ as the Query (Q) and the context sequence $\mathbf{F}_C$ as Key/Value (K/V):

$$\mathbf{v}_{\text{fused}} = \text{CrossAttention}(Q = \mathbf{v}_{BE}, K/V = \mathbf{F}_C) \quad (8)$$

In this configuration, $\mathbf{v}_{BE}$ queries the context $\mathbf{F}_C$ to align individual behavior and emotion with group dynamics, such as interaction with classmate. By using cross-attention, the model adjusts the behavior-emotion features based on group context, enhancing context-aware predictions. The resulting fused vector combines individual and group-level cues, leading to more accurate and context-sensitive engagement estimation.

V. EXPERIMENTS AND RESULTS

A. Experiments setup

All experiments were carried out on the Google Colab Pro platform using NVIDIA A100 and L4 GPUs (22.5 GB VRAM each). The proposed model and all comparative baselines are trained for 30 epochs with a batch size of 4, optimizing via AdamW with an initial learning rate of $1e-5$.

B. Results and Analysis

To evaluate the MoGLE-Net architecture on the HCMUE-SEGL dataset, we conducted two experimental groups. The first, an ablation study, tested unimodal configurations including RGB stream (Full-body action), Pose stream (Pose-based behavioral), Facial stream (Spatial-temporal facial emotion) to assess individual stream performance. The second compared multimodal strategies, including dual-stream (RGB+Facial, Pose+Facial) and MoGLE-Net (RGB + Pose + Facial + Context), to demonstrate the benefits of integrated signals for engagement estimation. Results are summarized in Table IV, with confusion matrices in Figure 3.

The RGB and Facial streams achieved accuracies of 86.00% and 73.00%, with F1-scores of 83.00% and 72.00%, respectively (Table IV). Their confusion matrices (Figures 3a and 3b) show misclassifications between "Normal" and "Engaged," and between "Engaged" and "Very Engaged" because some

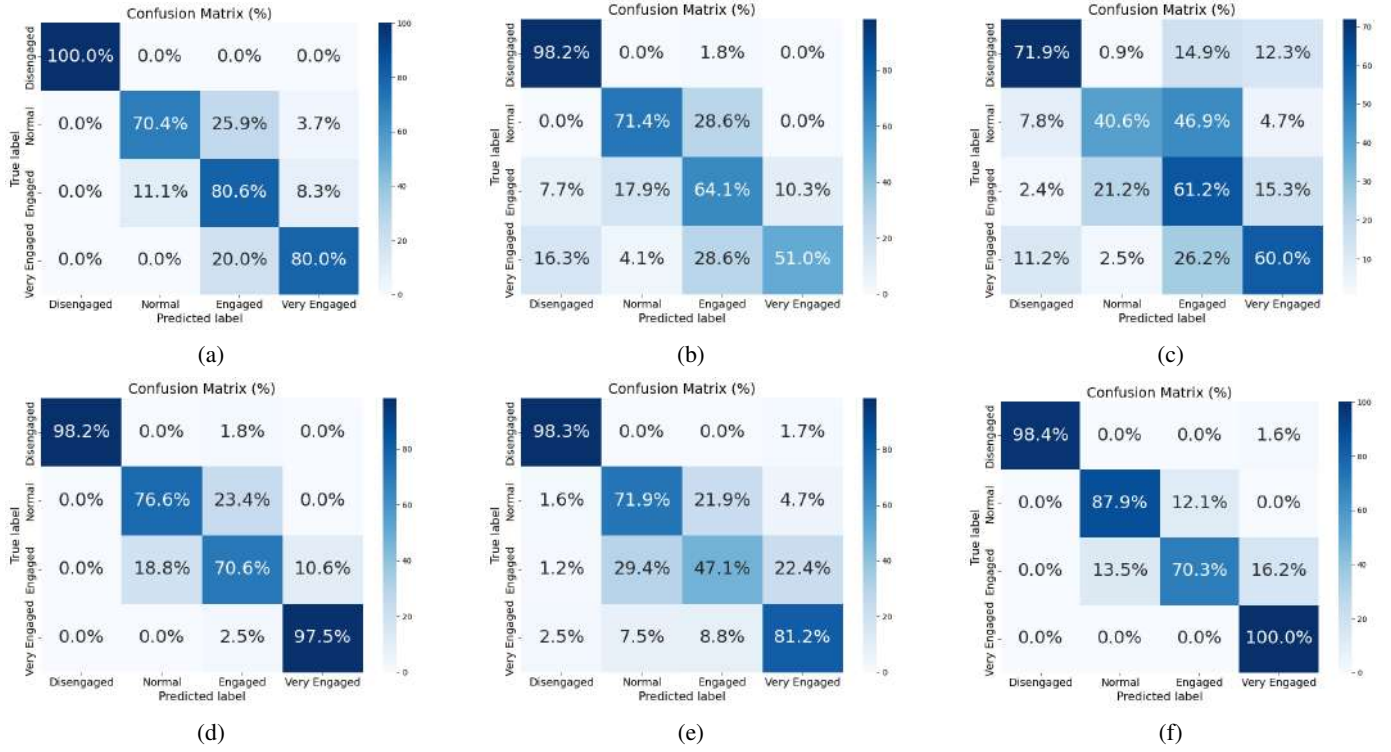


Fig. 3: Confusion matrices: (a) RGB Stream; (b) Facial Stream; (c) Pose Stream; (d) Action RGB + Facial Emotion; (e) Pose skeleton + Facial Emotion; (f) Action RGB + Pose skeleton + Facial Emotion + Context

behaviors and expressions appear in multiple engagement levels, making single-stream analysis insufficient. The Pose stream, with 60.64% accuracy and 58.92% F1-score (Figure 3c), struggles due to similar skeletal patterns in actions and lack of context.

TABLE IV: Performance comparison of different branches

Branch	F1	Prec.	Rec.	Acc.
RGB Stream	83.00%	84.00%	83.00%	86.00%
Facial Stream	72.00%	75.00%	72.00%	73.00%
Pose Stream	58.92%	60.97%	58.43%	60.64%
RGB Stream + Facial Stream	87.00%	87.00%	87.13%	87.13%
Pose Stream + Facial Stream	73.59%	73.73%	74.61%	76.74%
Action RGB + Pose Skeleton + Facial Emotion + Context	90.44%	90.83%	90.70%	90.70%

The two-modal configurations, Pose+Facial and RGB+Facial, improved performance over unimodal streams. Pose+Facial achieved 76.74% accuracy and 73.59% F1-score, surpassing Pose (60.64%, 58.92%) and Facial (73.00%, 72.00%). Similarly, RGB+Facial reached 87.13% accuracy and 87.00% F1-score, outperforming RGB (86.00%, 83.00%) and Facial alone. By combining skeletal or full-body motion with facial expressions, dual-modal models capture a holistic view of engagement. The synergy of behavioral cues and emotional signals effectively distinguishes subtle differences between levels like “Normal” and “Engaged,” unlike unimodal streams, which struggle with overlapping patterns. However, the confu-

sion matrix (Figure 3d and Figure 3e) shows misclassification between “Engaged” and “Normal” or “Very Engaged” due to the model’s focus on individual signals without capturing group interaction context, such as misclassifying a “happy” reader as “Engaged” instead of “Very Engaged” in a group discussion.

The full MoGLE-Net model (RGB+Pose+Facial+Context) achieved the highest performance with 90.70% accuracy and 90.44% F1-score, significantly reducing misclassification rates (Figure 3f). By integrating all modalities, it captures comprehensive behavioral, emotional, and contextual cues, effectively resolving ambiguities in engagement levels across unimodal and two-modal setups.

VI. CONCLUSION

This study proposed MoGLE-Net, a novel multimodal deep learning framework for assessing student engagement in group learning environments. By integrating behavioral, emotional, and contextual cues from various modalities, the model achieves an accuracy of 90.70%, demonstrating its effectiveness in predicting engagement levels. A HCMUE-SEGL dataset of 1,663 annotated video segments was constructed, enabling the model to process natural group learning interactions. The results highlight the advantages of using multimodal data, showing significant improvements over unimodal and two-modal approaches.

Despite these promising results, challenges remain when applying the model to real classroom data. Environmental factors such as lighting variation, occlusions, and spontaneous behaviors introduce noise and labeling ambiguity. In addition, generalization is limited: cultural differences shape how en-

gagement is expressed, and diverse classroom layouts alter behavioral and contextual cues, reducing robustness across educational settings.

In the future, the model will be expanded in applicability to diverse learning formats and integrate more granular facial feature analysis to improve sensitivity to subtle emotional states. Moreover, based on existing frameworks, future work should prioritize enhancing behavior recognition by moving beyond broad categories to capture micro-level behaviors and their social context. This involves developing models that can detect subtle actions like specific gestures, shifts in gaze direction, and purposeful fidgeting, which are more indicative of a student's cognitive state. Furthermore, it is crucial to analyze inter-student social dynamics to understand how behaviors influence one another, thereby capturing the essence of collaborative learning [30].

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